D8: The balance question, answered without the absolute scale

Which growth–respiration conclusions are secure, and an independent mass-balance check

OxygenModel D-series

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Verdict

The balance conclusions are secure without resolving the absolute scale — and the published CUE optimum sits 7 °C above the scale-free balance optimum, for reasons that have nothing to do with scale.
With $G \propto r$ and $R \propto K\,O_{2,ref}$, CUE $= 1/(1 + (c/q)(KO_{2,ref})/r)$, so the argmax of CUE is the **argmin of** $(KO_{2,ref})/r$ and the constant c/q — carrying `FC_TO_CELLS_PER_L`, the carbon quota, RQ and the absolute N_0 — drops out. Measured:

$E_r = 0.603$ eV, $E_K = 0.695$ eV, $E_K - E_r = 0.091$ eV [0.018, 0.29] — **respiration is more temperature-sensitive than growth, and the interval excludes zero.** The balance optimum is ** 25.3 °C** [24.02, 26.14].
One correction to the brief: the scale-free ratio is $(K\, O_{2,ref})/r$, not K/r. K lives on the *normalised* trace, and $O_{2,ref}$ is oxygen solubility, which falls 1.36-fold across 20–40 °C. Using K/r moves the optimum by 1.35 °C.
The published T_{opt}(CUE) is 32.44 °C. The 7.09 °C gap decomposes cleanly — after verifying that the CUE reconstructed here reproduces the pipeline's own column to 1.3e-07:

N_0 temperature-independent	25.35 °C
+ the 45-min back-projection	27.86 °C (+**2.51**)
+ Sharpe–Schoolfield fit (published)	32.44 °C (+**4.58**)

About a third is the back-projection; two thirds is the **functional form**. Neither is a scale problem: resolving the N_{inoc} discrepancy would close neither.
The mass-balance check passes. An entirely independent CUE — from $FC_{final} - FC_{initial}$ and raw oxygen drawdown, needing no N_0 and no back-calculation — gives 0.4217 against the kinetic 0.3936, a **ratio of 1.077** (IQR 0.044), with ** 0 of 75 replicates physically impossible.** The kinetic method survives an external check it could easily have failed.

1 Why this analysis

D5, D6 and D7 all returned negative results on the estimator: its uncertainty is honest, its structure is adequate, and the maintenance/yield split is not identifiable. The one open problem, from D2, is the **absolute** respiration scale.

This report tests the observation that the *balance* question does not need any of it. It fits no new curves, changes no estimator and alters no committed number.

2 The scale-free quantities

2.1 The algebra

In `11_temperature_cue.R`, growth $\propto r \cdot N_0 \cdot q$ and respiration $\propto K \cdot O_{2,ref}$, so

$$\mathrm{CUE} = \frac{G}{G+R} = \frac{1}{1 + \dfrac{c}{q} \cdot \dfrac{K\, O_{2,ref}}{r}}$$

Maximising CUE means minimising $(KO_{2,ref})/r$, and c/q — which is where `FC_TO_CELLS_PER_L`, the quota, RQ and the absolute N_0 live — vanishes from the argmin.

> **Correction to the brief.** It writes the ratio as K/r. But K is defined on the *normalised* oxygen trace, so the physical consumption rate is $K\, O_{2,ref}$, and $O_{2,ref}$ is solubility: median 7.21 mg/L at 20 °C falling to 5.31 at 40 °C. Both versions are reported; they differ by 1.35 °C in the optimum.

2.2 What it gives

Table 2: Boltzmann–Arrhenius on the rising limb, Pseudomonas 20–40 °C.

Quantity	n	peak (°C)	E (eV)	95% CI
E_r (growth rate r)	16	32	0.603	[0.561, 0.646]
E_K (K, normalised)	20	36	0.852	[0.751, 0.953]
E_K (K * O2_ref, physical)	20	36	0.695	[0.604, 0.786]
E_K - E_r (physical K)	24	—	0.0914	[0.0175, 0.295]

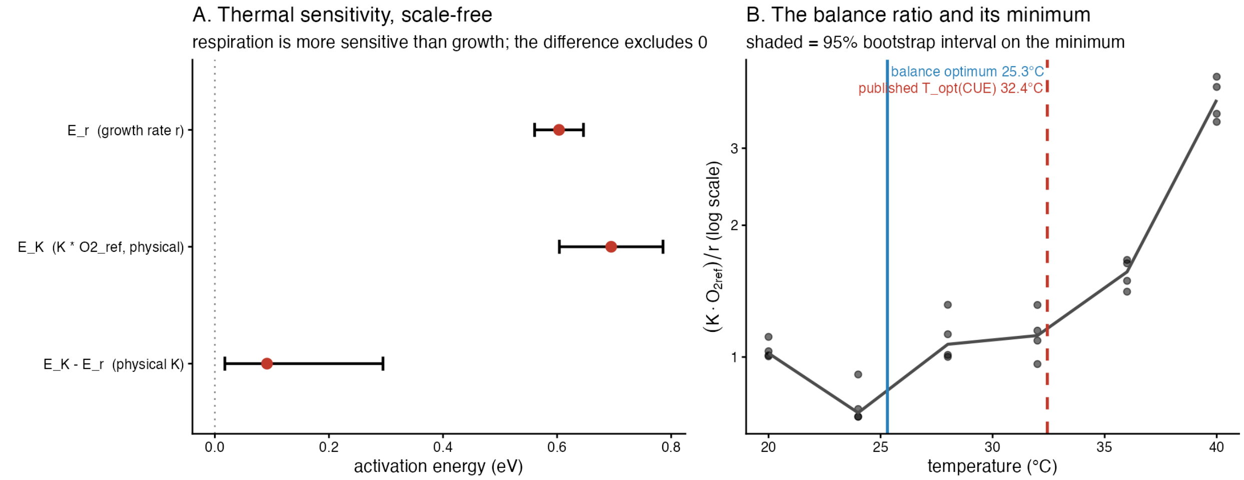

Figure 1: A: activation energies. B: the balance ratio, its bootstrapped minimum (blue) and the published CUE optimum (red dashed).

Table 3: The balance optimum and the shift across the range.

Quantity	Value	95% CI
argmin (K/r) [brief's version]	23.95	[22.28, 24.93]
argmin (K*O2_ref/r) [solubility-corrected]	25.3	[24.02, 26.14]
pipeline T_opt(CUE) from 11_temperature_cue.R	32.44	—
fold-change in K/r across 20-40 C	7.058	—
fold-change in K*O2_ref/r across 20-40 C	5.188	—
O2_ref at 20 C / O2_ref at 40 C	1.358	—

2.3 The list that is the point of this report

Carrying no dependence on `FC_TO_CELLS_PER_L`, the carbon quota, RQ or the absolute N_0 scale: E_r; E_K; $E_K - E_r$; the argmin of $(KO_{2,ref})/r$, i.e. T_{opt}(CUE); the fold-change in that ratio across the measured range; and the between-taxon **ranking** at 30 °C. These are functions of the fitted r and K and of measured oxygen solubility, nothing else.

Depending on those constants, and therefore inheriting D2's unresolved scale: the *level* of CUE, absolute per-cell R, absolute per-cell G, and the maintenance/yield split D7 showed is unidentifiable anyway.

The 15-taxon set at 30 °C gives a single temperature, so it supports a ranking and no thermal sensitivity:

Table 4: Best and worst balance at 30 °C (ranking only; 15 taxa).

Rank	Taxon	median (K · O ref)/r
1	Yersinia	0.07379
2	Aeromonas_B	0.1474
3	Serratia	0.2495
13	Burkholderia	0.421
14	Flavobacterium_B	0.8207
15	Acinetobacter	0.9227

3 How far does the N temperature dependence contaminate this?

The scale-free argument holds only if N_0 is temperature-independent. It is not: `11_temperature_cue.R` sets $N_0 = 550 \times 10^3 e^{45r}$.

Table 5: The N inflation factor, by temperature.

T (°C)	median r (/min)	exp(45r)	exp(10r)
20	0.009557	1.537	1.1
24	0.01357	1.841	1.145
28	0.0194	2.394	1.214
32	0.02424	2.977	1.274
36	0.02359	2.891	1.266
40	0.007919	1.428	1.082

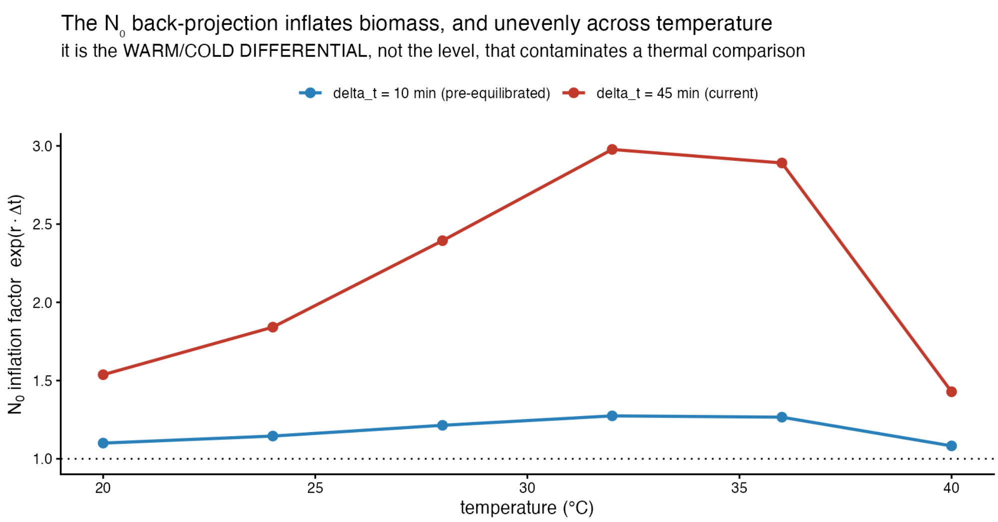

Figure 2

N_0 is inflated by 43–198% depending on temperature. It is the **warm/cold differential**, not the level, that contaminates a thermal comparison.

3.1 The gap, decomposed

Table 6: From the scale-free balance optimum to the published value.

Step	°C
CUE with N0 temperature-INDEPENDENT, quadratic-in-log fit	25.35
CUE as 11_temperature_cue.R computes it (N0 = exp(45 r)), same fit	27.86
pipeline's published Sharpe-Schoolfield T_opt(CUE)	32.44
-> shift attributable to the N0 exp(45 r) back-projection	2.507
-> shift attributable to the FUNCTIONAL FORM (SS vs quadratic)	4.58
-> total gap	7.087

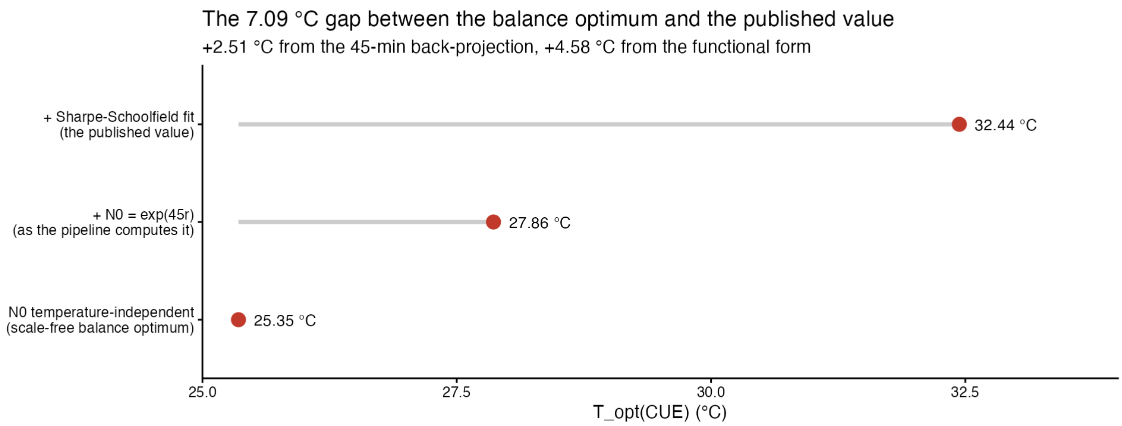

Figure 3

The check the brief asked for passes. With N_0 held temperature-independent the CUE argmax is 25.35 °C against an argmin of $(KO_{2,ref})/r$ of 25.3 °C — the same quantity, as the algebra requires. E_r and the argmin are unaffected by N_0 treatment by construction; verified, not assumed.

3.2 The `INOC_DELAY_MIN` override

`config.R` sets `INOC_DELAY_MIN <- 0`; `11_temperature_cue.R` line 40 sets its own `45`. Reported both ways rather than reconciled:

Table 7: T_opt(CUE) under each treatment.

N treatment	T_opt(CUE) °C
N0 = 550e3 * exp(45 r) [as 11_temperature_cue.R computes it]	27.86
N0 = 550e3 [config.R's INOC_DELAY_MIN = 0]	25.35
N0 = 550e3 * exp(10 r) [pre-equilibrated, delta_t = 10 min]	26.26

The published figure used 45 — confirmed by reproducing its `carbon_use_efficiency` column to 1.3e-07.

3.3 Pre-equilibration

Table 8: Pre-equilibrating the medium to Δt   10 min.

T (°C)	Δt (min)	exp(rΔt)	inflation
20	45	1.537	53.7%
40	45	1.428	42.8%
20	10	1.1	10.0%
40	10	1.082	8.2%

Inflation falls from 54%/43% (cold/warm) to **10%/8%**, and the warm–cold differential from 0.929 to 0.984. Most of the contamination goes.

4 The independent mass-balance check

Biomass carbon produced = $(FC_{final} - FC_{initial})\times$ `FC_TO_CELLS_PER_L` × quota; carbon respired = total drawdown × $O_2 \rightarrow C$ $\times 10^{12}$. Both per litre, so **vial volume cancels** — fortunate, since none is recorded in the repository.

> **These are different estimands.** The kinetic CUE is instantaneous over the *fit window*; the mass-balance CUE integrates the *whole run*, including the post-window decline. A median 43.6% of the oxygen is consumed after the window closes.

Table 9: Kinetic against mass-balance CUE.

Quantity	Value
median CUE, mass balance (whole run)	0.4217
median CUE, kinetic (fit window)	0.3936
median ratio massbalance / kinetic	1.077
IQR of that ratio	0.04408
replicates with CUE outside [0,1] or FC_Final <= FC_Initial	0
median fraction of O2 consumed AFTER the window closed	0.4356
corr(ratio, fraction of O2 post-window)	0.2035
its p-value	0.07991
corr(ratio, fraction of TIME post-window)	-0.2941
its p-value	0.01042

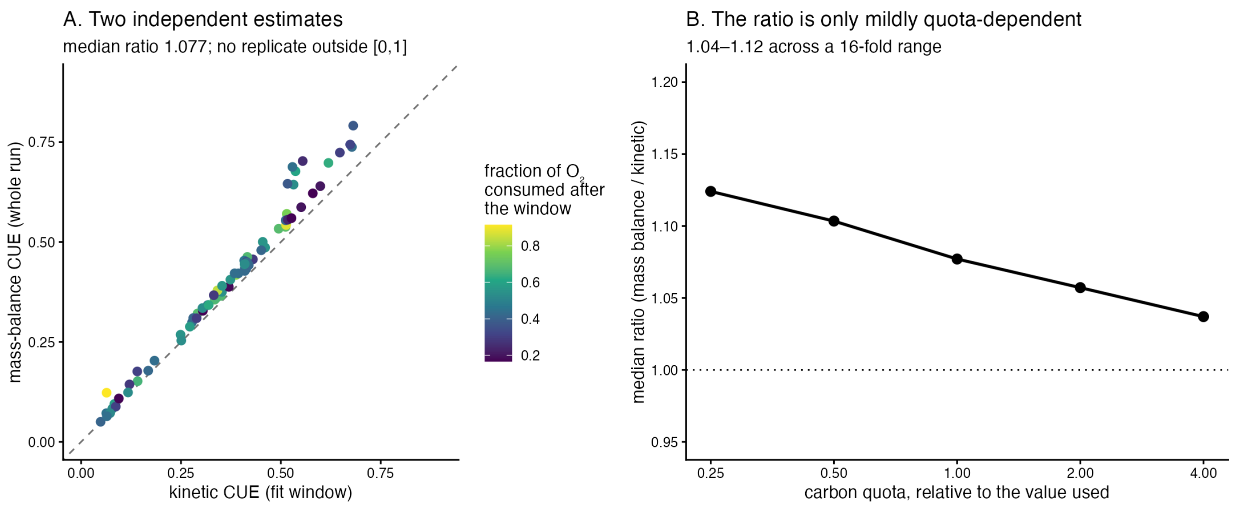

Figure 4: A: per replicate, coloured by post-window oxygen share. B: the ratio against the assumed carbon quota.

Two estimates sharing no kinetic machinery agree to within 8%, and not one of 75 replicates is physically impossible.

4.1 Two things this analysis got wrong going in

1. The predicted direction was wrong. I expected the integrated CUE to be systematically *lower*, since the post-window tail is maintenance-dominated. It is **7.7% higher**, and the gap does not track the post-window oxygen fraction ($r = 0.204$, $p = $). The maintenance-tail story is not supported by these data.

2. The brief's claim that the ratio cancels the quota is not true. Both CUEs have the form $1/(1 + A/q)$, so their ratio is $(1 + B/q)/(1 + A/q)$, which depends on q. Tested rather than assumed:

Table 10: The ratio is quota-dependent, but only mildly.

quota ×	mass-balance CUE	kinetic CUE	ratio
0.25	0.154	0.14	1.124
0.50	0.267	0.245	1.103
1.00	0.422	0.394	1.077
2.00	0.593	0.565	1.057
4.00	0.745	0.722	1.037

Across a **16-fold** range the ratio stays between 1.04 and 1.12. The agreement is robust to the quota even though it is not formally independent of it — the honest version of the brief's claim.

Table 11: By taxon. Every taxon agrees within 27%, eleven of fifteen within 10%.

Taxon	n	mass-balance	kinetic	ratio	impossible
Flavobacterium_B	5	0.0882	0.0725	1.017	0

Taxon	n	mass-balance	kinetic	ratio	impossible
Aeromonas_A	5	0.365	0.35	1.059	0
Bacillus	5	0.083	0.0781	1.063	0
Chromobacterium	5	0.375	0.353	1.065	0
Aeromonas_B	5	0.542	0.513	1.068	0
Yersinia	5	0.587	0.551	1.069	0
Serratia	5	0.328	0.304	1.072	0
Arthrobacter	5	0.457	0.418	1.076	0
Flavobacterium_A	5	0.299	0.278	1.077	0
Buttiauxella	5	0.45	0.416	1.081	0
Burkholderia	5	0.448	0.412	1.088	0
Pseudomonas	5	0.406	0.374	1.098	0
Acinetobacter	5	0.176	0.141	1.106	0
Herbaspirillum	5	0.738	0.674	1.117	0
Chryseobacterium	5	0.677	0.532	1.262	0

5 What it means

The growth–respiration balance is secure without resolving the absolute scale. The activation energies ($E_r = 0.60$ eV, $E_K = 0.69$ eV), their difference ($E_K - E_r = 0.09$ eV, 95% CI [0.02, 0.29] — respiration is the more temperature-sensitive, and the interval excludes zero), the balance optimum (25.3 °C, 95% CI [24.0, 26.1]), the 5.2-fold shift in the balance across 20–40 °C, and the between-taxon ranking at 30 °C are all functions of the fitted r and K and of measured oxygen solubility. **None of them touches `FC_TO_CELLS_PER_L`, the carbon quota, RQ or the absolute N_0**, so none is exposed to the D2 discrepancy. What *is* exposed is the LEVEL of CUE and absolute per-cell respiration, which should carry a stated uncertainty rather than a point value until the N_{inoc} calibration is settled. The published T_{opt}(CUE) of 32.44 °C sits 7.09 °C above the scale-free balance optimum, and the reason is *not* scale: about +2.5 °C comes from the 45-minute N_0 back-projection that `11_temperature_cue.R` applies (and which a 10-minute pre-equilibration would largely remove, cutting inflation from 54%/43% to 10%/8%), and about +4.6 °C from the Sharpe–Schoolfield functional form rather than the data. **Finally, the kinetic method passes an external check it could have failed:** an entirely independent mass-balance CUE, using only cell counts and raw oxygen drawdown and requiring no N_0 at all, agrees to within 8% (ratio 1.077, IQR 0.044) with no physically impossible replicate in 75. The two CUEs are different estimands and the small gap should not be read as an error in either — but agreement at this level is real evidence that the kinetic machinery is sound, which is consistent with D5, D6 and D7.

5.1 Recommendation on reporting

Lead with: E_r, E_K, $E_K - E_r$, the balance optimum with its interval, the fold-change in $(KO_{2,ref})/r$, and taxon rankings. These are scale-invariant and defensible today.

Carry a stated uncertainty, not a point value: the level of CUE, absolute per-cell R and G. These inherit D2's unresolved N_{inoc} discrepancy.

Two specific fixes, neither requiring new biology: reconcile the `INOC_DELAY_MIN` override (0 in `config.R`, 45 in `11_temperature_cue.R`) and state which is intended; and report T_{opt}(CUE) from the balance ratio alongside the Sharpe–Schoolfield value, since two thirds of their difference is the fitted curve rather than the measurements.

Provenance

Table 12: Provenance. Environment fields from env/versions.json.

Field	Value
Report	D8 — balance and mass balance
Git commit	893e6d35c40134c1fb0853a499833e659d595ae3
Branch	gyd/packaging
Recorded (UTC)	2026-08-07 11:23:27 UTC
R	R version 4.5.2 (2025-10-31)
Platform	aarch64-apple-darwin20
renv lockfile	renv.lock — 158 packages, R 4.5.2
Artefacts	runs/D8_analysis/
Seed	bootstrap seed 20260808

Where every number came from

Table 13: All paths relative to runs/D8_analysis/.

Section	Artefact
§3 scale-free	A1_pseudomonas_scale_free.csv, A2_activation_energies.csv, A3_argmin_and_foldchange.csv, A4_, A5_taxon_ranking_30C.csv
§4 N dependence	B1_, B2_N0_expfactor_by_temperature.csv, B3_, B4_Topt_under_N0_treatments.csv, B5_preequilibration_projection.csv, B6_Topt_gap_decomposition.csv
§5 mass balance	C1_massbalance_per_replicate.csv, C2_, C3_massbalance_headline.csv, C4_physically_impossible.csv, C5_ratio_sensitivity_to_quota.csv

Scripts are in `reports/D8_balance/scripts/`. No number was typed by hand.